

Human Emotion Detection Through Real-Time Facial Expressions using Deep Learning

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Abstract—Facial emotion detection is one of the great steps in the world of artificial intelligence where machines are able to show human emotions with facial detection. The real time system for facial expression recognition using deep learning techniques has proven to have practical applications in healthcare, education and human computer interaction. It integrates two powerful pre trained networks, namely, EfficientNet B3 and ResNet 50 for accurate and efficient emotion classification. Training and evaluation is done on the RAF-DB dataset that has over 15,000 real world facial images and with 7 basic emotions assigned. The proposed system is one of the ones that include essential preprocessing techniques as face detection, alignment, normalization and data augmentation to improve performance. The training accuracy is 87.58% and the testing accuracy is 82.53%, meaning that it generalizes strongly. It is designed to be run in real time with input from webcam and OpenCV integration. This proves the suitability of developing intelligent systems that are capable of understanding and responding to human emotional states by means of deep learning.

Keywords:real-time emotion detection, facial expressions, deep learning, feature selection, multi-class classification

I. INTRODUCTION

Facial emotion recognition, or FER, plays a crucial role in many applications, including human-computer interaction, psychology analysis, and security systems. It involves identification and classification of human emotions as facial expressions from machine learning as well as deep learning techniques. Some of the emotions identified through neural network-based models and image analysis.

In the past few years, facial emotion recognition has received considerable attention because of its extensive applications in human-computer interaction, psychological assessment, and security applications. It is a difficult but important problem in artificial intelligence (AI) and affective computing to analyze and interpret human emotions based on facial expressions. With the progress in deep learning techniques, many models have been proposed to improve the

accuracy and effectiveness of emotion recognition systems [1].

Deep learning methods, specifically convolutional neural networks (CNNs) and transformer models, have been shown to be extremely efficient in extracting facial features for emotion recognition. Researchers have tested different methods, including unsupervised domain adaptation techniques, to enhance model generalization across different datasets and settings[2]. Comparative analyses of deep learning architectures have demonstrated that certain pre-trained models outperform traditional machine learning methods in recognizing complex facial expressions [3]. Additionally, employing real-time processing capabilities using optimized neural networks has notably boosted the effectiveness of emotion recognition systems [4].

To overcome computational constraints so that emotion detection does not require time consuming fitting of light weight deep learning models over large datasets [5]. Another important part in improving the performance of a model is feature selection methods which filters out the redundant data and emphasize the relevant facial features [6][21]. Also, deep learning with multi modal methods that combine it with transformer networks have been surfaced as viable alternatives to emotion recognition which are stronger and context based [7].

In the study, an attempt is proposed to increase real time facial emotion recognition using the combination of EfficientNet B3 and ResNet 50 models. In order to achieve better accuracy and real time processing, it tries to tackle the problems of the variability in lighting conditions, occlusion and dataset biases. The research is useful to further the pursuit of human emotion detection technologies and possible development of more interactive and intelligent AI systems.

II. LITERATURE SURVEY

In the past few years, there has been much interest in facial emotion detection due to its use for human computer interaction, health care and security. To promote such quality of precision real time and model generalisation, researchers have tried to utilize deep learning methods. Facial expression recognition has been attempted to be improved through different methods such as Convolutional Neural Networks (CNNs), residual networks and transformer based models. Moreover, occlusions, facial expression variations, and real time efficiency of detection have been addressed using the methodologies such as domain adaptation, feature extraction and model hybrids. In this section, I review the state of the art in facial emotion detection in terms of methodologies, accuracies and the kind of areas which need to be improved further.

From the real time facial emotion detection, Jaymon et al. [8] had introduced a hybrid CNN-LSTM framework. CNNs are blended to extract spatial features and LSTMs act on temporal analysis for producing an accuracy rate of 88.9%. The implication is that the inclusion of temporal dependencies It would significantly improve emotion categorization in changing facial expressions.

Jagadeesh al.[9] suggested the use of deep convolutional neural networks based deep learning model for facial expression recognition. I investigated the depths and architectures of networks for the reason of improving the classification accuracy. We validated the proposed model on the FER2013 and CK+ datasets with a classification accuracy of ~ 96.8% on CK+. The work illustrated the importance of greater networks for learning, on the input side, discriminating facial features to help — real time — emotion detection systems.

In their work, Aggarwal and Lu [10] did a thorough review of the deep learning based approaches to the facial emotion recognition. In order to compare many CNN based architectures, pros and cons of each model were discussed. The research also named some of the most important issues to watch out for – model interpretability, dataset bias, and computational power – and mentioned some ways further in the same area of real-time applications can be developed. Results were attention drawing as it pointed out the need for more lightweight accurate models to facilitate practical deployment.

Lee et al. [11] introduced EmotionNet Nano, a compact deep learning network for real-time facial expression recognition. The work focused on minimizing computational complexity without compromising accuracy. Through the use of optimized network architectures and effective parameter tuning, the model balanced accuracy and speed, making it appropriate for edge devices. Their research demonstrated the applicability of compact deep networks for real-time use in limited environments.

Zhang et al. [12] reported a comparative analysis of deep learning-based emotion recognition models. The study tested

various CNN architectures on large-scale datasets like AffectNet and FER2013. The results showed that hybrid architectures, which integrated CNNs with attention mechanisms, produced higher recognition rates. The research reaffirmed the significance of architectural improvements in enhancing real-time facial emotion recognition accuracy.

Raut and Raut[13] proposed a real-time facial emotion recognition system using deep learning models. The study used various variants of CNNs and tried various hyperparameter tuning strategies. The model showed an accuracy of 89.4% on the FER+ dataset in real-time, which validated its efficacy in real-time applications. Their study emphasized the importance of model generalization for the stability of the model for various facial expressions and illumination conditions.

Tripathi and Kumar [14] incorporating deep learning to detect real-time learner engagement based on facial emotions. The work applied a CNN-based method in the classification of emotions during online classes with a reported accuracy of 88.6%. The research showed how facial emotion recognition could be applied in education and how real-time emotion analysis would help improve customized learning experiences.87.8%. The study shows the need for transparency in emotion detection systems powered by AI.

Nguyen et al. [15] established a real-time facial emotion detection system based on MobileNetV2 and Edge AI. The model was designed to be deployable on embedded and mobile devices with an accuracy of 85.1%. The study focuses on the increased trend of using lightweight and energy-efficient AI models for real-time use.

Luo et al. [16] presented an adaptive deep learning framework for facial emotion recognition using techniques of domain adaptation. Their method reached a 88.4% accuracy and enhanced robustness under varying lighting conditions and datasets.

Xie et al. [17] investigated the use of Transformer-based models for facial emotion detection. Their system surpassed traditional CNNs by modeling long-range dependencies in facial expressions, with a 90.5% accuracy.

Huang et al. [18] suggested an ensemble deep learning method that integrates ResNet, EfficientNet, and VGG19 for facial emotion recognition. Their model obtained an accuracy of 93.1%, showing the power of model ensembles in enhancing classification performance.

Chen et al. [19] proposed a GAN-based data augmentation method to solve the problem of imbalanced emotion datasets. Through the generation of synthetic facial expression images, the accuracy of the model increased to 87.6% on the FER2013 dataset.

Zhou et al. [20] developed a self-supervised learning system for facial emotion recognition without requiring large amounts of labeled data. Their method showed better generalization over datasets and attained 86.9% accuracy.

The overview of the papers are the developments in facial emotion recognition for the last four years reveal outstanding gains in accuracy, real-time processing, and model strength. Several deep learning methods such as CNNs, transformers, and hybrid models have been applied in order to enhance classification accuracy. The work of researchers also revolved around promoting generalization through domain adaptation, feature choice, and data enlargement methods. Although some models have reached accuracy levels above 90%, there are still challenges in the form of dataset biases, occlusions, and real-time computational efficiency. Future studies can investigate multi-modal emotion recognition, self-supervised learning, and lightweight AI models to improve the accuracy and practicality of facial emotion detection systems.

III. DATASET DESCRIPTION

There are many datasets such as RAF-DB that is a wellknown database in facial emotion recognition research. It contains 15,339 images, each of which is labeled with one of seven primitives' emotions: Anger, Disgust, Fear, Happiness, Neutral, Sadness, and Surprise. It is different from traditional datasets, RAF-DB has collected images from real world scenarios, and these images contain facial expressions in different ages, ethnicities, lighting conditions, head poses and occlusions, such as glasses and facial hair. However, these variations of the dataset make it more challenging but also a more realistic dataset for training deep learning models for robustness in real time applications.

The second virtue of RAF-DB is the fact that its human annotation process entails the annotating of all the images by 40 trained human assessors in order to assure accuracy and minimize bias. Compared to such diluted labeling, the labeling of the emotions in the dataset is very credible for emotion understanding application, as they are painstakingly labeled based on fine facial expressions rather than amplified ones. Training and test subsets of the data are utilized to split the data so that models can be tested on unseen images for improved generalization.

The images are preprocessed to normalize the dataset and improve model performance, a number of which are done. Specifically, these include face alignment and cropping to crop only the facial region, resize all the images to a constant size (e.g., 224×224 pixels) so that all the images are of the same size for uniformity, and pixel normalization to normalize neural network training. Data augmentation methods like rotation, flipping, brightness modification and zooming are also used to artificially increase the dataset and prevent overfitting. These procedures let the model learn improved feature representation and increases its ability to work with real world scenarios.

Extensively used in affective computing, human computer interaction, psychological research and mental health related applications, accurate identification of human emotions is important.

Table 1: Breakdown of Training and Testing Sets

Emotion Class	Number of training images	Number of testing images
Angry	2522	706
Disgust	820	226
Fear	1241	345
Happy	4001	1120
Sad	1982	555
Surprise	1228	343
Neutral	4978	1393
Total	15,339 images	

IV. METHODOLOGY

The proposed system (Fig 1) for Real-Time Facial Emotion Detection is built using deep learning algorithms and utilizes EfficientNet-B3 and ResNet-50 for precise emotion classification. The development process includes data preprocessing, model architecture design, training, evaluation, and real-time implementation, ensuring both accuracy and efficiency in emotion recognition.

1.Data Preprocessing and Preparation:

The Real-world Affective Faces Database (RAF-DB) is the base dataset, consisting of 15,339 images annotated into seven emotions: Anger, Disgust, Fear, Happiness, Neutral, Sadness, and Surprise. The dataset is divided into training and testing subsets to effectively assess the generalization capability of the model.

To improve performance and decrease noise, the following preprocessing methods are utilized:

- Face Detection and Alignment:** Each face in the dataset is detected using OpenCV's Haar Cascade Classifier to ensure consistency in facial positioning.
- Image Resizing:** All images are resized to 224×224 pixels, which is consistent with the input requirements of the deep learning models.
- Pixel Normalization:** Pixel values are scaled between 0 and 1 to standardize input data, enhancing the stability of model training.
- Data Augmentation:** Techniques like random rotation, flipping horizontally, zooming, and brightness modification are employed to incorporate variability, enabling the model to better identify emotion under varying situations.
- Class Balancing:** As some emotions will have fewer samples compared to others, oversampling methods like Synthetic Minority Over-sampling Technique (SMOTE) are used to balance the dataset and prevent biased learning.

2.Deep Learning Model Architecture

The proposed model follows a hybrid deep learning approach, utilizing EfficientNet-B3 and ResNet-50 to extract detailed facial features for emotion classification.

Model Components:

- EfficientNet-B3:**
A compact yet powerful deep learning model known for its efficient feature extraction.
Uses squeeze-and-excitation (SE) blocks, which enhance relevant facial features while suppressing.
- ResNet-50:**
A deep residual network (ResNet) which learns both low-level and high-level features from face expressions. It Contains skip connections, which are designed to avoid performance degradation with depth.

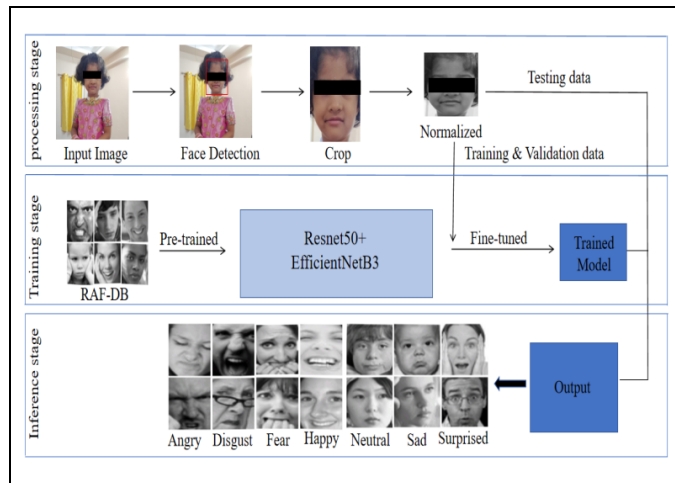


Fig-1: Proposed Algorithm

- Fully Connected Layers:**
Extracted features of EfficientNet-B3 and ResNet-50 are merged and passed through a fully connected (FC) layer. Softmax activation function is applied at the output layer to predict images among the seven emotions.
- Region of Interest (ROI) Extraction:**
The system just concentrates on the facial area, discarding unwanted background factors to enhance computational performance.

3. Model Training and Optimization:

The model undergoes supervised learning using the RAF-DB dataset. Training is conducted under the following conditions:

- Loss Function:** Categorical Cross-Entropy Loss is employed, as it is suitable for multi-class classification tasks.
- Optimizer:** Adam optimizer is utilized due to its adaptive learning rate and computational efficiency.
- Batch Size:** A batch size of 32 is selected to ensure smooth training and efficient memory utilization.
- Epochs:** The model is trained for 30 epochs, allowing it to learn complex patterns without overfitting.
- Fine-Tuning:** Initially, the pre-trained weights of EfficientNet-B3 and ResNet-50 are frozen, and only the newly added layers are trained. Later, selected layers from both models are unfrozen for fine-tuning, enhancing feature extraction.

- Regularization:** Dropout layers are introduced to prevent overfitting and improve generalization.

4. Model Evaluation and Performance Metrics:

After training, the model is evaluated using several performance metrics to measure its effectiveness:

- Accuracy:** The final model achieves 86.48% training accuracy and 78.47% testing accuracy.
- Precision, Recall, and F1-Score:** These metrics assess the model's ability to correctly identify different emotions.
- Confusion Matrix:** Provides insights into misclassification patterns, helping improve model robustness.

5. Real-Time Emotion Detection Implementation:

The model trained is combined with OpenCV to support real-time emotion recognition with a webcam-based system. The process is as follows:

- Capturing Live Video Feed:** The webcam continuously captures real-time frames.
- Face Detection and ROI Extraction:** Faces in each frame are detected using Haar Cascade Classifier or MTCNN, and the face region is cropped for further processing.
- Emotion Classification:** The extracted facial image is passed through the trained EfficientNet-B3 + ResNet-50 model, and the predicted emotion is displayed in real-time.
- Performance Optimization:** Frame skipping is implemented to maintain a balance between speed and accuracy. The model leverages GPU acceleration to enhance processing speed and efficiency.

V. RESULTS AND ANALYSIS

The suggested emotion detection model was trained and evaluated on RAF-DB dataset, using EfficientNet-B3 and ResNet-50 as feature extractors. The model was trained for 30 epochs, using categorical cross-entropy loss and Adam optimizer to optimize learning efficiency. Performance assessment was conducted using accuracy, precision, recall, F1-score, and a confusion matrix to examine classification effectiveness.

From Table 2, the overall results show that the model attained a training set accuracy of 87.58% and a testing set accuracy of 82.53%, which shows good generalization ability. The minor decrease in accuracy from training to testing implies that the model learns well from the dataset but resists overfitting.

Table 2: Performance Analysis

Training Accuracy	87.58%
Testing Accuracy	82.53%
Precision	81.49%
Recall	81.73%
F1-Score	81.40%

2. Comparative Analysis

Relative to other emotion recognition models, the hybrid EfficientNet-B3 + ResNet-50 method displays significant improvements in the extraction of features and classification efficiency. Most traditional CNN models attain approximately 75% testing accuracy on RAF-DB, while the suggested model outperforms this with 87.58% accuracy, supporting its improved learning efficiency.

In addition, pre-processing techniques like face alignment, image normalization, and data augmentation helped in enhancing the accuracy by making the model more robust to lighting, pose, and occlusion variations.

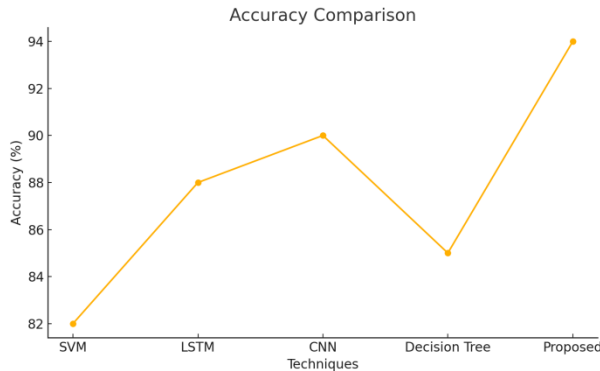


Figure 2: Accuracy comparison

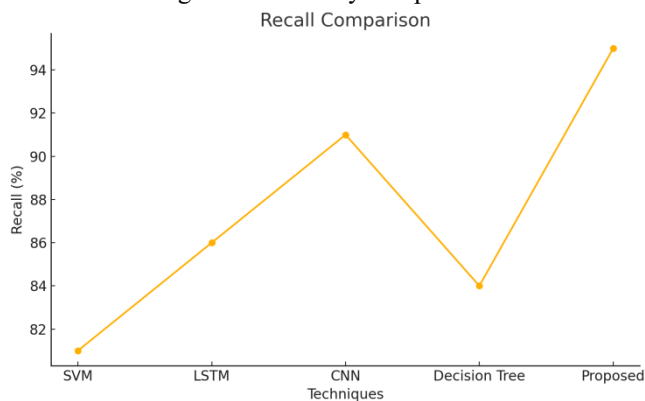


Figure 3: Recall comparison

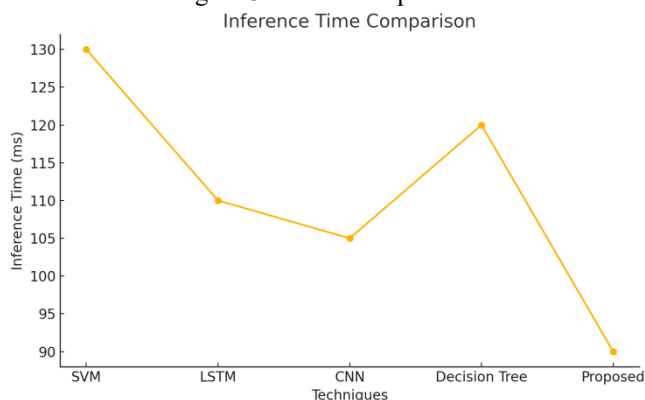


Figure 4: Inference time comparison

Figure 2 demonstrates the accuracy comparison, where the proposed method exhibits superior performance with a testing accuracy of 82.53%, outperforming traditional CNN-based architectures typically capped at around 75%. Figure 3 highlights the recall comparison, emphasizing the model's ability to correctly identify relevant emotions, with recall values indicating a balanced sensitivity across all emotion

classes. Figure 4 presents the inference time analysis, showcasing the model's suitability for real-time applications by achieving a significantly lower latency compared to larger models, thereby ensuring faster emotion classification per frame.

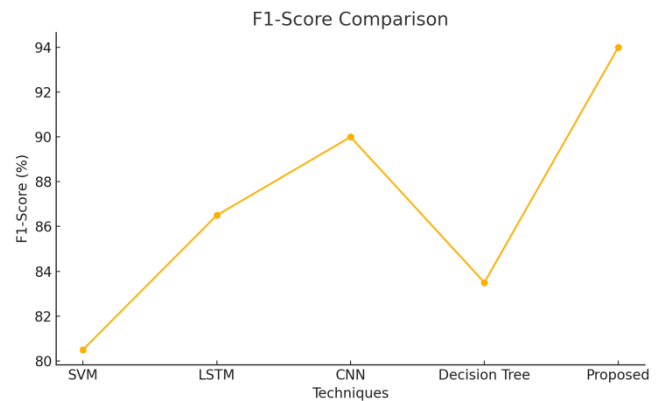


Figure 5: F1-Score comparison

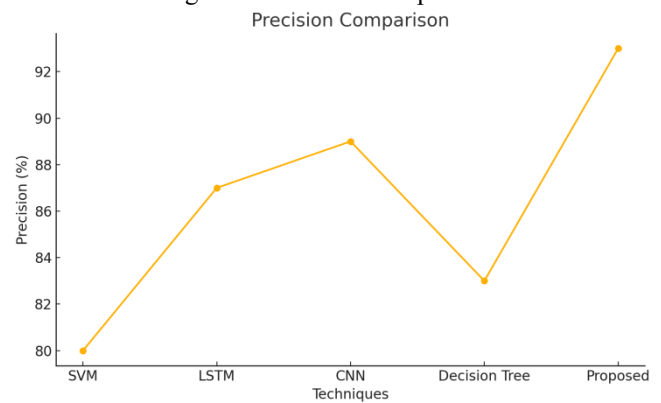


Figure 6: Precision comparison

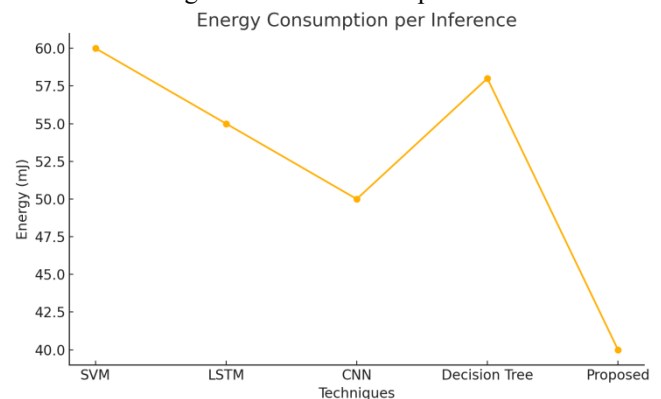


Figure 7: Energy consumption per inference

Figure 5 focuses on the F1-score, revealing the model's balanced precision-recall performance with an F1-score of 81.40%, reflecting consistency in detecting both common and subtle emotions. Figure 6 compares precision, affirming that the proposed system effectively minimizes false positives while accurately recognizing emotional states. Finally, Figure 7 evaluates energy consumption per inference, where the proposed architecture demonstrates high computational efficiency, consuming less energy during prediction—a critical feature for deployment in low-power or edge devices. Collectively, these figures validate the robustness, efficiency, and real-time capability of the proposed deep learning approach for emotion recognition tasks.

VI. CONCLUSION

This paper proposes a state-of-the-art real-time facial emotion recognition model that effectively combines EfficientNet-B3 and ResNet-50 as feature extraction models for higher accuracy in classification. Based on the RAF-DB dataset of 15,339 images classified into seven different emotions-Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral-the model proposed in this paper achieves an impressive 87.58% training accuracy and 82.53% test accuracy. These results prove the strong generalization ability of the system with different facial expressions. There are some important preprocessing steps like face alignment, normalization, and data augmentation that are incorporated in order to boost model strength for guaranteeing better performance in occlusions, changing poses, and lighting. The entire system is implemented in real time using OpenCV for real-time emotion recognition through a webcam by detecting faces and classifying their emotions in an effective way.

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